Privacy-Preserving Multicenter Evaluation of ICU Clinical Decision Support: A Federated Framework Across Eight U.S. Health Systems

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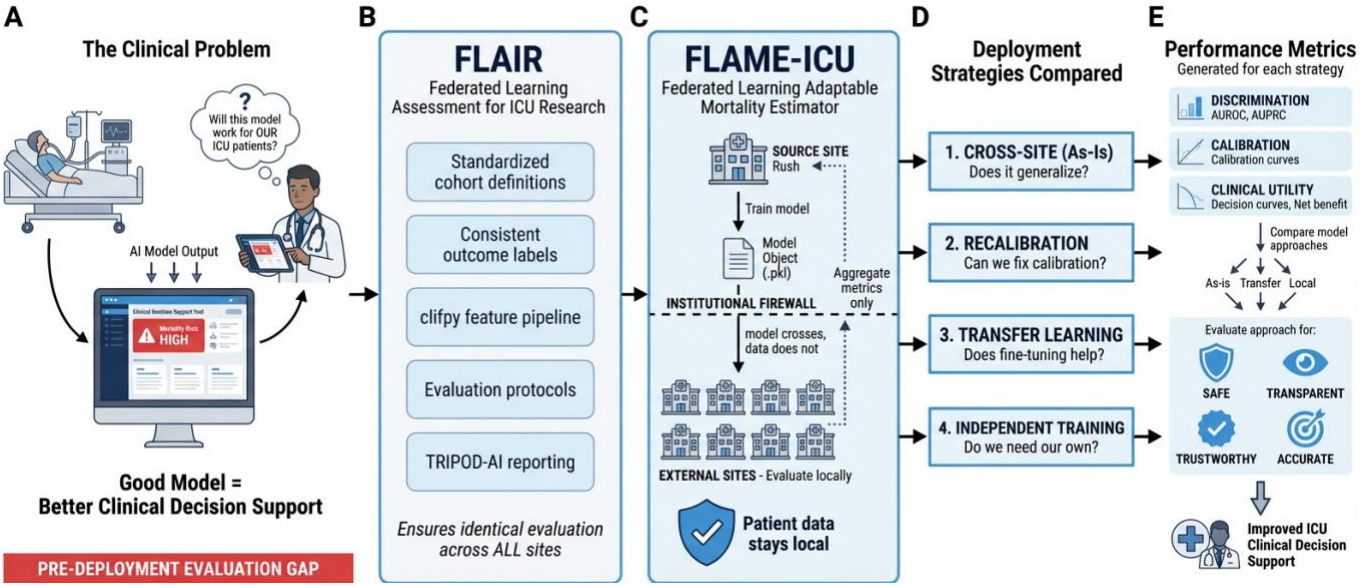
BACKGROUND: Deploying machine learning models across health systems remains challenging, with widely implemented algorithms, such as commercial sepsis prediction models, demonstrating poor external validity[1]. While prediction models can support clinical decision-making in the Intensive Care Unit (ICU), external validation typically requires sharing patient-level data, creating regulatory and data privacy barriers that impede rigorous evaluation before deployment[2]. Using inpatient mortality prediction for ICU patients, we compared three deployment strategies across U.S. health systems to guide clinical decision support implementation.

METHODS: We developed FLAIR (Federated Learning Assessment for ICU Research), a privacy-preserving benchmarking framework that enables evaluation of ICU prediction models without data sharing (**Figure 1**)[3]. Using FLAIR, we conducted FLAME (Federated Learning Assessment for Mortality Evaluation), a federated analysis across the Common Longitudinal ICU data Format (CLIF) Consortium—a diverse network of 8 U.S. academic medical centers[4–6]. We included adult ICU patients admitted between 2018–2024 who survived at least 24 hours. Demographics, vital signs, laboratory values, and respiratory support data were extracted at 24 hours post-ICU admission. The primary outcome was in-hospital mortality. Using inpatient mortality prediction as an initial use case, we compared three deployment strategies: (1) cross-site validation using the lead site (Rush) model on 2024 data, (2) transfer learning with local fine-tuning on 2018–2023 data and 2024 testing, and (3) independent site-specific training on 2018–2023 data with 2024 testing. We trained XGBoost and ElasticNet classifiers and distributed model objects to 7 external sites for local evaluation[7,8].

RESULTS: The cohort comprised 372,673 ICU encounters across 8 U.S. health systems (mean age 62 years, 45% female) (**Table 1**). In-hospital mortality ranged from 11% to 18% across sites. Demographics varied: the proportion Black ranged from 2.4% to 59.1%, and the proportion Hispanic from 2.1% to 18.8%. For the XGBoost model, cross-site validation AUC was 0.85 [range 0.78–0.89], transfer learning AUC 0.86 [range 0.79–0.89], and independent training AUC 0.87 [range 0.85–0.89] (**Table 2**). Brier scores ranged from 0.15–0.19 for cross-site validation, improving to 0.13–0.16 with independent training. The poorest cross-site validation site saw the largest gains: XGBoost AUC rose from 0.78 to 0.85 with independent training, and ElasticNet AUC increased from 0.77 to 0.82 with transfer learning alone.

CONCLUSIONS: For in-hospital mortality prediction, site-specific model training outperformed Rush models, with transfer learning providing meaningful improvements when local retraining is not feasible. CLIF addresses the data harmonization challenge by providing a common structure for ICU data, while FLAIR enables federated model evaluation without sharing patient data. Currently, this framework is available to institutions that have adopted the CLIF standard. However, the availability of MIMIC-IV in CLIF format provides an open-source entry point for broader adoption and enables benchmarking against private institutional data without direct data sharing[6]. Together, these tools support rigorous, transparent evaluation of clinical AI prior to deployment and are extensible to additional ICU prediction tasks.

Figure 1: FLAIR and FLAME-ICU: Standardized Framework for Federated ICU Mortality Model Development and Evaluation



(A) The clinical problem: AI-based clinical decision support tools trained at one institution may not generalize to others, creating a pre-deployment evaluation gap. **(B)** The Federated Learning Assessment for ICU Research (FLAIR) framework establishes standardized cohort definitions, outcome labels, feature pipelines using clifpy (the Python package for the Common Longitudinal ICU data Format), evaluation protocols, and TRIPOD-AI reporting to ensure consistent evaluation across participating sites. **(C)** The Federated Learning Adaptable Mortality Estimator for the ICU (FLAME-ICU) architecture enables model sharing while maintaining patient privacy; the trained model object crosses the institutional firewall, but only aggregate performance metrics are returned—patient data remains local. **(D)** Four deployment strategies compared: cross-site validation (as-is), recalibration, transfer learning with local fine-tuning, and independent local training. **(E)** Performance metrics generated for each strategy include discrimination (AUROC, AUPRC), calibration curves, and clinical utility (decision curves, net benefit), evaluated against criteria for safe, transparent, trustworthy, and accurate clinical decision support.

Abbreviations: *AI, artificial intelligence; AUPRC, area under the precision-recall curve; AUROC, area under the receiver operating characteristic curve; CLIF, Common Longitudinal ICU data Format; ICU, intensive care unit; TRIPOD-AI, Transparent Reporting of a Multivariable Prediction Model for Individual Prognosis or Diagnosis–Artificial Intelligence.*

Table 1: Baseline Characteristics - In-Hospital Mortality

Variable	HS1	HS2	HS3	HS4	HS5	HS6	HS7	HS8
Cohort								
Total	69,830	37,892	69,334	55,957	21,292	35,473	29,857	53,038
Demographics								
Age, mean (SD)	63.7 (16.8)	61.2 (16.8)	61.7 (17.3)	65.2 (16.4)	61.4 (17.0)	60.6 (16.9)	58.7 (17.5)	62.9 (17.1)
Sex								
Male	39,125 (56.0)	19,917 (52.6)	37,267 (53.8)	31,979 (57.1)	12,619 (59.3)	18,454 (52.0)	16,814 (56.3)	29,214 (55.1)
Female	30,694 (44.0)	17,970 (47.4)	32,060 (46.2)	23,978 (42.9)	8,673 (40.7)	17,019 (48.0)	13,043 (43.7)	23,812 (44.9)
Race								
White	51,222 (73.3)	13,890 (36.7)	40,298 (58.1)	38,311 (68.5)	17,828 (83.7)	14,253 (40.2)	8,577 (28.7)	43,814 (82.6)
Black	9,269 (13.3)	19,549 (51.6)	20,803 (30.0)	5,831 (10.4)	515 (2.4)	13,597 (38.3)	17,658 (59.1)	3,748 (7.1)
Asian	2,480 (3.5)	1,438 (3.8)	2,995 (4.3)	1,672 (3.0)	566 (2.7)	1,106 (3.1)	574 (1.9)	3,102 (5.8)
Other	4,730 (6.8)	919 (2.4)	4,531 (6.5)	3,822 (6.8)	262 (1.2)	5,810 (16.4)	693 (2.3)	225 (0.4)
Ethnicity								
Hispanic	6,631 (9.5)	1,527 (4.0)	3,187 (4.6)	2,182 (3.9)	1,548 (7.3)	6,655 (18.8)	1,813 (6.1)	1,098 (2.1)
Vital Signs								
Heart Rate Max	103.4 (22.1)	105.0 (22.4)	105.9 (22.0)	103.8 (20.7)	108.0 (24.6)	105.4 (21.8)	107.7 (21.4)	105.1 (22.6)
Heart Rate Min	69.5 (15.6)	72.2 (16.9)	70.4 (16.1)	70.2 (15.4)	72.0 (16.0)	70.1 (15.7)	73.1 (16.9)	68.6 (15.3)
SBP Min	92.9 (19.4)	95.3 (21.6)	93.3 (20.2)	90.5 (16.5)	92.7 (17.1)	98.0 (18.9)	92.7 (19.2)	90.9 (21.3)
MAP Median	83.0 (12.1)	87.2 (13.9)	84.7 (13.1)	77.4 (11.0)	82.7 (12.0)	87.2 (13.0)	69.9 (26.8)	84.2 (12.7)
Respiratory Rate Max	29.9 (9.1)	31.9 (9.2)	31.4 (8.7)	28.1 (6.3)	27.0 (6.5)	28.5 (6.5)	27.3 (6.7)	30.2 (9.8)
SpO2 Min	91.1 (5.7)	90.8 (7.0)	91.4 (6.1)	91.8 (5.2)	90.8 (5.1)	91.2 (5.7)	90.8 (6.6)	89.5 (6.5)
Temperature Max	37.4 (0.7)	37.3 (0.7)	37.4 (0.7)	37.4 (0.7)	37.4 (0.7)	37.1 (0.8)	37.2 (0.7)	37.4 (0.7)
Laboratory Values								
Lactate Max	2.9 (2.8)	3.1 (3.1)	3.1 (3.0)	2.7 (2.2)	3.1 (2.8)	2.7 (2.7)	3.3 (3.2)	3.2 (3.2)
Creatinine Max	1.6 (1.8)	2.1 (2.5)	1.7 (2.1)	1.5 (1.6)	1.6 (2.0)	2.1 (2.6)	1.9 (2.1)	1.5 (1.6)
BUN Max	27.6 (22.8)	30.3 (25.8)	26.3 (23.1)	27.7 (22.9)	28.4 (24.2)	28.8 (24.5)	28.5 (23.9)	27.1 (21.8)
Bilirubin Max	1.4 (3.0)	1.5 (3.5)	1.2 (2.6)	1.9 (4.3)	1.9 (3.8)	1.5 (3.6)	1.7 (3.9)	1.5 (3.1)
Platelet Min	196.4 (99.5)	204.5 (104.8)	198.0 (97.2)	191.8 (103.5)	186.9 (98.8)	203.1 (102.3)	194.5 (102.4)	191.1 (95.4)
WBC Max	12.7 (10.0)	12.8 (11.5)	13.3 (9.1)	10.1 (6.8)	16.0 (26.6)	17.0 (29.2)	13.2 (11.6)	12.9 (9.7)
Albumin Min	3.1 (0.6)	3.0 (0.7)	3.1 (0.7)	3.1 (0.7)	2.8 (0.7)	2.8 (0.7)	3.2 (0.7)	2.8 (0.7)
Clinical Assessments								
GCS Total	13.5 (3.0)	13.4 (3.0)	13.3 (3.1)	14.6 (1.4)	13.4 (3.0)	13.4 (3.1)	12.7 (3.6)	12.9 (3.4)
SOFA Score								
SOFA Total	4.7 (4.2)	3.9 (3.4)	4.8 (4.1)	4.8 (3.4)	4.6 (3.9)	3.9 (3.8)	5.4 (4.6)	4.8 (4.3)
Medications								
Vasopressors Given	22,910 (32.8)	11,350 (30.0)	20,166 (29.1)	17,872 (31.9)	7,942 (37.3)	6,455 (18.2)	9,298 (31.1)	16,883 (31.8)
Outcomes								
Hospital Mortality								
Train	7,863 (13.7)	2,273 (11.0)	8,672 (14.5)	5,402 (13.8)	2,380 (13.0)	3,634 (12.0)	4,439 (17.5)	5,989 (13.1)
Test	1,684 (13.8)	1,877 (10.9)	1,528 (15.8)	2,329 (13.9)	416 (13.7)	592 (11.3)	796 (17.8)	1,004 (13.5)

Abbreviations: HS = Health System; SBP = Systolic Blood Pressure; MAP = Mean Arterial Pressure; BUN = Blood Urea Nitrogen; WBC = White Blood Cell; GCS = Glasgow Coma Scale; SOFA = Sequential Organ Failure Assessment; SpO2 = Oxygen Saturation. Train: 2018–2023; Test: 2024.

Table 2: In-Hospital Mortality Prediction Metrics

Experiment / Metric	Health System 1		Health System 2		Health System 3		Health System 4		Health System 5		Health System 6		Health System 7	
	XGB[a]	EN[a]	XGB[a]	EN[a]	XGB[a]	EN[a]	XGB[a]	EN[a]	XGB[a]	EN[a]	XGB[a]	EN[a]	XGB[a]	EN[a]
External Validation[b]														
AUROC	0.88 (0.87-0.89)	0.86 (0.85-0.87)	0.86 (0.85-0.86)	0.84 (0.83-0.85)	0.88 (0.87-0.89)	0.86 (0.85-0.87)	0.78 (0.77-0.79)	0.77 (0.76-0.78)	0.84 (0.82-0.86)	0.83 (0.81-0.85)	**0.89 (0.88-0.90)**	0.87 (0.86-0.89)	0.85 (0.84-0.86)	0.84 (0.82-0.85)
Brier	0.15 (0.15-0.16)	0.16 (0.16-0.17)	0.18 (0.18-0.18)	0.17 (0.17-0.17)	0.15 (0.15-0.16)	0.17 (0.16-0.17)	0.17 (0.17-0.17)	0.16 (0.15-0.16)	0.19 (0.18-0.19)	**0.14 (0.13-0.15)**	0.17 (0.16-0.17)	0.18 (0.17-0.18)	0.16 (0.15-0.16)	0.19 (0.19-0.20)
NB@20%[e]	0.00	0.00	-0.05	-0.03	0.02	0.03	-0.02	-0.01	-0.03	0.01	0.04	**0.05**	-0.00	-0.02
Transfer Learning[c]														
AUROC	0.88 (0.87-0.89)	0.88 (0.87-0.89)	0.86 (0.85-0.87)	0.85 (0.84-0.86)	0.88 (0.87-0.89)	0.87 (0.87-0.88)	0.79 (0.78-0.80)	0.82 (0.81-0.83)	0.85 (0.83-0.86)	0.84 (0.82-0.86)	**0.89 (0.88-0.90)**	0.88 (0.87-0.89)	0.85 (0.84-0.87)	0.85 (0.83-0.86)
Brier	0.15 (0.15-0.15)	0.14 (0.14-0.15)	0.18 (0.17-0.18)	0.16 (0.15-0.16)	0.15 (0.15-0.16)	**0.14 (0.13-0.14)**	0.17 (0.17-0.17)	0.17 (0.17-0.18)	0.18 (0.18-0.19)	0.15 (0.14-0.15)	0.16 (0.16-0.17)	0.14 (0.13-0.15)	0.15 (0.15-0.16)	0.15 (0.15-0.16)
NB@20%[e]	0.01	0.02	-0.05	-0.03	0.03	0.05	-0.02	-0.01	-0.03	0.00	0.04	**0.07**	-0.00	0.00
Independent[d]														
AUROC	0.89 (0.88-0.90)	0.88 (0.87-0.89)	0.87 (0.86-0.87)	0.85 (0.84-0.86)	0.88 (0.88-0.89)	0.87 (0.87-0.88)	0.85 (0.84-0.85)	0.82 (0.81-0.83)	0.85 (0.83-0.87)	0.84 (0.82-0.86)	**0.89 (0.88-0.90)**	0.88 (0.87-0.89)	0.87 (0.86-0.88)	0.85 (0.83-0.86)
Brier	0.14 (0.13-0.14)	0.14 (0.14-0.15)	0.14 (0.14-0.14)	0.16 (0.15-0.16)	0.13 (0.13-0.14)	0.14 (0.13-0.14)	0.16 (0.16-0.16)	0.17 (0.17-0.18)	0.16 (0.15-0.17)	0.15 (0.14-0.15)	**0.13 (0.13-0.14)**	0.14 (0.13-0.15)	0.14 (0.14-0.15)	0.15 (0.15-0.16)
NB@20%[e]	0.03	0.02	-0.01	-0.03	0.05	0.05	-0.00	-0.01	-0.00	0.00	**0.07**	0.07	0.01	0.00

[a] XGB: XGBoost; EN: ElasticNet (logistic regression with elastic net regularization).

[b] External Validation: Rush-trained model evaluated directly on external sites (Rush internal: AUROC=0.879, Brier=0.129).

[c] Transfer Learning: Rush model weights used to initialize site-specific model training.

[d] Independent: Models trained and evaluated entirely on site-specific data.

[e] NB@20%: Net benefit at 20% decision threshold, quantifying clinical utility by weighing true positives against false positives at a clinically relevant risk threshold.

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